

Structure of the variable
parent_child_hierarchy_with_boundaries
containing hierarchy of parent-child interfaces

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```
parent_cluster_idx → dict [11 keys]
  parent_cluster_id → int
  parent_phase_id → int
  parent_n_pixels → int
  n_children → int
  children → list [1 items]
    [0] → dict [6 keys]
      child_cluster_id → int
      child_phase_id → int
      child_n_pixels → int
      relationship → str: 'enclosed'
      interface → dict [3 keys]
        n_boundary_pixels → int
        parent_side → dict [9 keys]
          indices → array[8] <int32>
          coords → array[8×2] <float64>
          X → array[8] <float32>
          Y → array[8] <float32>
          q_mean → array[4] <float64>
          M_mean → array[3×3] <float64>
          q_best_cluster → array[8×4] <float64>
          M_best_cluster → array[8×3×3] <float64>
          n_pixels → int
        child_side → dict [9 keys]
          indices → array[8] <int32>
          coords → array[8×2] <float64>
          X → array[8] <float32>
          Y → array[8] <float32>
          q_mean → array[4] <float64>
          M_mean → array[3×3] <float64>
          q_best_cluster → array[8×4] <float64>
          M_best_cluster → array[8×3×3] <float64>
          n_pixels → int
      is_lamellar → int
  parent_boundary_indices_all → array[8] <int64>
```

```
child_boundary_indices_all → array[8] <int64>  
parent_boundary_mask → array[64493] <bool>  
child_boundary_mask → array[64493] <bool>  
parent_boundary_avgori_all → array[64493×3×3] <float64>  
child_boundary_avgori_all → array[64493×3×3] <float64>
```